Jerby-Arnon 2021 Synovial Sarcoma — summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

3 patients (SyS12pt/13/14) intermix on one continuous manifold; the kNN graph is a single component across the θ sweep (same-sample frac 1.00→0.72 at θ=5). Cross-platform concat fails (10x/SmartSeq2 stay platform-segregated, ~0.87 at every θ), so the run is scoped to the 10x leaf.

Gate 2 — module geography → GO (non-branching)

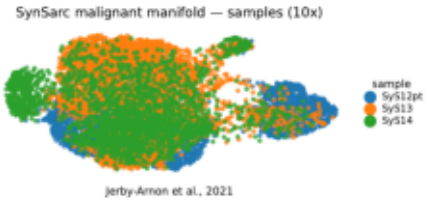
Gast & pSom co-vary (smoothed r=+0.76) while Crest opposes both (−0.75 / −0.97); the Crest-high decile sits at the trajectory terminus (mean pt 1.00). pSom-dominant → Linear-pSom/Crest.

Gate 3 — hub cross-repression → NA

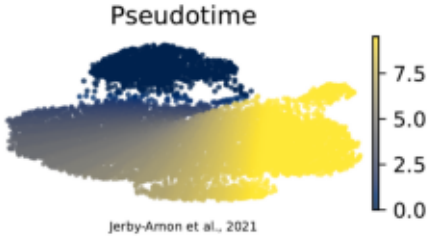
Single monotone curve (0 forks), so the branching-only cross-repression test does not apply.

STEP FIGURES

Gate 1: integration



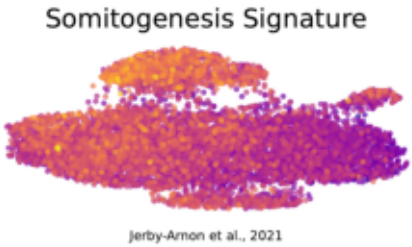
Trajectory / root



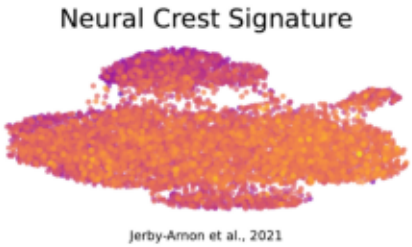
Gast



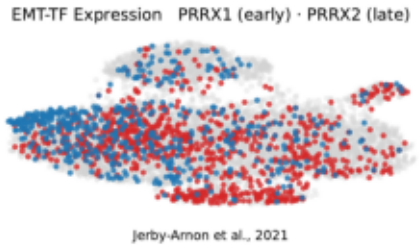
pSom



Crest



EMT-TFs



Dataset. Synovial sarcoma (Jerby-Arnon et al., 2021); 10x scRNA-seq (UMI counts); 8,322 malignant cells, 3 patients, after island removal. The SmartSeq2 leaf (4,371 cells, 11 patients) reproduces the same axis direction as an independent check.

Verdict. ✓ Pattern reproduces (non-branching). A single gastrulation/somitogenesis-high origin flows to a neural-crest terminus; the EMT axis runs PRRX1 (root) → PRRX2 (terminus). G1 GO / G2 GO / G3 NA — the Linear-pSom/Crest outcome shared by most CCCA datasets.

Parameter choices

Min cells/sample	none (3 balanced samples; ratio 1.59)	**Integration**	Harmony θ=5 (θ=10 over-integrates)
Scope	10x leaf (platform un-integratable)	**Trajectory**	scFates curve N=20 λ=0.3 μ=100 (SA 2.58)
Root node	tip 2 = Gast/pSom-high, Crest-low	**EMT-TFs**	PRRX1 (root) → PRRX2 (terminus)